Chi-Yao Huang

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Education

- **Arizona State University (ASU)** — Tempe, AZ
  Ph.D. in Computer Science — *Current*
- **Arizona State University (ASU)** — Tempe, AZ
  M.S. in Robotics and Autonomous Systems (AI Track); GPA: 4.00/4.00 — *Aug 2021 – May 2023*
- **National Taiwan University (NTU)** — Taipei, Taiwan
  M.S. in Mechanical Engineering; GPA: 3.83/4.3 — *Sep 2015 – Jun 2017*
- **National Sun Yat-Sen University (NSYSU)** — Kaohsiung, Taiwan
  B.S. in Mechanical and Electro-Mechanical Engineering — *Sep 2010 – Jun 2014*

Publications

- **Domain Expansion: A Latent Space Construction Framework for Multi-Task Learning (ICLR 2026):** Developed "Domain Expansion," a multi-task learning framework to address latent representation collapse caused by conflicting gradients. Designed an orthogonal pooling mechanism that separates objectives into mutually orthogonal subspaces to reduce interference. Validated the framework on ShapeNet and MPIIGaze, showing an interpretable and compositional latent space. (project | paper | code | talk)
- **VOCAL: Visual Odometry via ContrAstive Learning (WACV 2026):** Introduced "VOCAL," a visual odometry framework that reformulates state estimation as a label-ranking problem. Combined Bayesian inference with contrastive learning to align visual features with camera states. Evaluated on the KITTI dataset, showing multimodal compatibility and explainable spatial intelligence. (project | paper | talk | demo)

Work Experience

- **Arizona State University (ASU)** — Tempe, AZ
  Ph.D. Student – Prof. Yezhou Yang — *Aug 2021 – Present*
  - **Research:** Developing a latent-centric framework that unifies 3D geometry, semantics, and generative modeling, bridging the gap between the real world and AI.
  - **Funding:** Supported by Toyota Research Institute of North America (TRINA).
- **Advanced and Creative Team, HTC VIVE (Acquired by Google)** — New Taipei, Taiwan
  Core Member, Team Lambda (VR/AR Innovations) — *Sep 2017 – Feb 2021*
  - **Core Contributions:** Served as a core team member in design reviews, technical planning, and engineering execution for VR/AR products. Mentored junior engineers and contributed to key technical decisions in tracking and spatial computing systems.
  - **VIVE COSMOS:** Developed a multi-camera VR tracking system with 0.4 mm accuracy using multi-camera bundle adjustment, and designed a mothership SLAM system for VR products. (product | demo)
  - **VIVE FOCUS 3:** Led a team to prototype an MR system integrating gesture interaction and tightly coupled visual-inertial SLAM, achieving trajectory jitter below 0.1 mm and a 4× speed-up on embedded systems. (product)
  - **VIVE XR Elite:** Implemented low-latency visual-inertial odometry for XR applications. Designed and prototyped core data structures and the processing pipeline for a mixed-reality system. Developed an obstacle avoidance system to improve user safety in mixed-reality environments. (product)
  - **VIVE FLOW:** Designed an inside-out tracking algorithm for limited-FoV sensors. Developed a prototype motion-tracking pipeline integrating real-time hand tracking. Collaborated with hardware and firmware teams to resolve CPU load and thermal issues in an AR device. (product)
  - **Scene:** Engineered a voxel-based obstacle mapping system for VR safety and integrated semantic maps for real-world interaction.
- **NTU Robotics Lab, National Taiwan University** — Taipei, Taiwan
  Graduate Student and Vice System Manager – Prof. Han-Pang Huang — *Sep 2015 – Jun 2017*
  - **Vice System Manager:** Developed orientation materials and managed robotic equipment including robotic arms and mobile robots.

- **Research:** Conducted research in SLAM, 3D reconstruction, dense mapping, and robot motion planning.

- **Company of Air Defense Artillery Battalion, R.O.C. Army** — Taiwan
  Battalion Dispatcher — *Sep 2014 – Sep 2015*
  - **Operations:** Managed the dispatch and coordination of military vehicles, trucks, and anti-aircraft systems during training and military exercises.

Projects

- **Aerial Pose Estimation (Sponsored by Toyota Research Institute of North America (TRINA)):** Developed high-altitude pose estimation for aerial robots using IR and IMU sensors under varying weather conditions. (demo)
- **3D Landmark Reconstruction for Robot Localization and Mapping:** Used learning-based 3D reconstruction and semantic labeling to jointly optimize robot trajectories and object poses. (demo)
- **3D Reconstruction and Path Planning with Signed Distance Function:** Fused optical flow with feature points for efficient SLAM, modified bundle adjustment using Lie groups and quaternions, implemented voxel hashing for dense mapping, and integrated SLAM with biped robot path planning.
- **Humanoid Robot:** Designed ZMP trajectories to enable biped locomotion on uneven terrain. (demo)
- **Commercial Vehicle Speed Control System:** Designed a PID controller for vehicle speed regulation.
- **Stairlift for Elderly People:** Developed a PLC-based stair-climbing system to improve mobility for elderly users.

Research Areas

- Representation Learning, Spatial Intelligence, SLAM, Visual Odometry, 3D Computer Vision, Robotics

Honors, Awards & Grants

- **Research Grant, Toyota Research Institute of North America (TRINA) (2024 – Present)**
- **Ira A. Fulton Schools of Engineering Fulton Fellows Award, ASU**
- **Student with Distinction, Arizona State University**

Skills

- **Programming Languages:** Python, C/C++, JavaScript
- **Libraries & Tools:** PyTorch, OpenCV, Eigen, Ceres, Keras, OpenGL, SIMD, CEVA, Hexagon

Languages

- English (Fluent), Mandarin (Native)